Tianzhe Chu

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Research Highlights

Nested Sparsity for Scalable Pretraining and Continual Learning
A transformer architecture that nests conditional computation in attention with MoE sparsity, with μP, Muon, and AttnRes enabled. At equal compute and sparsity, Nestor improves pretraining, adapts to new domains with less forgetting, and extends to longer contexts more effectively than conventional MoEs.
[paper] [website] [code]

Benchmarking Dexterous Embodied System
A real-world benchmark probing whether embodied agents can perceive the table, track the evolving game state across turns, and act on it through dexterous manipulation, with Texas Hold'em as a rich physical testbed. As co-leader, I led the embodied agent design and the perception benchmark measuring how reliably agents recover world state from raw observation.
[paper] [website] [code]

SFT Memorizes, RL Generalizes
Under equal training compute, RL with verified rewards generalizes to unseen textual and visual variants while SFT memorizes the training distribution, though SFT remains a necessary format-stabilization step before RL. Further shows that scaling up verification steps during RL training boosts its generalization.
[paper] [website] [code]

Experience

Advanced Machine Intelligence Singapore
Research Intern, World Model Pretraining *Sep 2026 - Now*

Hyperbolic San Francisco, CA, US
Research Intern, Agent and AI4Math *Dec 2025 – Feb 2026*

University of California, Berkeley Berkeley, CA, US
Visiting Undergraduate, EECS & BAIR *Aug 2022 – May 2023*

Education

The University of Hong Kong Hong Kong SAR
Ph.D. candidate at IDS advised by Prof. Yi Ma *Sep 2024 – Present*

ShanghaiTech University Shanghai, China
B.Eng in Computer Science and Technology (Honor) *Sep 2020 – Jun 2024*

Publications

(* denotes equal contribution)

Nested Sparsity for Scalable Pretraining and Continual Learning
Tianzhe Chu*, Druv Pai*, Yi Ma
Preprint [paper] [website] [code]

SFT Memorizes, RL Generalizes: A Comprehensive Study of Foundation Model Post-training
Tianzhe Chu*, Yuexiang Zhai*, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, Yi Ma
ICML 2025 [paper] [website] [code]

Pointer-CAD: Unifying B-Rep and Command Sequences via Pointer-based Edges & Faces Selection
Dacheng Qi, Chenyu Wang, Jingwei Xu, **Tianzhe Chu**, Zibo Zhao, Wen Liu, Wenrui Ding, Yi Ma, Shenghua Gao
CVPR 2026 [paper]

Seeing from Another Perspective: Evaluating Multi-View Understanding in MLLMs
Chun-Hsiao Yeh*, Chenyu Wang*, Shengbang Tong, Ta-Ying Cheng, Ruoyu Wang, **Tianzhe Chu**, Yuexiang Zhai, Yubei Chen, Shenghua Gao, Yi Ma
AAAI 2026 [paper] [website] [code]

White-Box Transformers via Sparse Rate Reduction: Compression Is All There Is?
Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Hao Bai, Yuexiang Zhai, Benjamin D. Haeffele, Yi Ma
JMLR [paper]

Image Clustering via the Principle of Rate Reduction in the Age of Pretrained Models
Tianzhe Chu*, Shengbang Tong*, Tianjiao Ding*, Xili Dai, Benjamin D. Haeffele, René Vidal, Yi Ma
ICLR 2024 [paper] [code]

Emergence of Segmentation with Minimalistic White-Box Transformers
Yaodong Yu*, **Tianzhe Chu***, Shengbang Tong, Ziyang Wu, Druv Pai, Sam Buchanan, Yi Ma
CPAL 2024 (Oral), NeurIPS 2023 XAI Workshop (Oral) [paper] [website] [code]

White-Box Transformers via Sparse Rate Reduction
Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Benjamin D. Haeffele, Yi Ma
NeurIPS 2023 [paper] [website] [code]

DexHoldem: Playing Texas Hold'em with Dexterous Embodied System
Feng Chen*, **Tianzhe Chu***, Li Sun*, Pei Zhou*, Zhuxiu Xu, Shenghua Gao, Yuexiang Zhai, Yanchao Yang, Yi Ma
Preprint [paper] [website] [code]

GenDexHand: Generative Simulation for Dexterous Hands
Feng Chen*, Zhuxiu Xu*, **Tianzhe Chu**, Xunzhe Zhou, Li Sun, Zewen Wu, Shenghua Gao, Zhongyu Li, Yanchao Yang, Yi Ma
Preprint [paper]

GAUSS Eval: Human–LLM Judge Consistency Analysis
Tianzhe Chu*, Jiaxin Zhang*, Zhenyu Liao, Qiuyu Ren, Tahsin Saffat, Zitong Yang, Yi Ma, Yue Zhang*
Blog Post [website]

Open Source

Deep Representation Learning Book — Core Contributor [code]
Open-source textbook on the principles of deep representation learning.

CRATE — Core Contributor [code]
White-box transformer architecture derived from sparse rate reduction.

Selected Awards

Outstanding Graduate — Shanghai, China
ShanghaiTech University — *Jul 2024*

Merit Student — Shanghai, China
ShanghaiTech University — *Dec 2023*

Invited Talks

- Localize the Credits of Reasoning in Roaring 2025: LLM Reasoning Paper Potpourri @ ALICE, Aug. 2025
- SFT Memorizes, RL Generalizes @ BAAI, Jun. 2025
- SFT Memorizes, RL Generalizes @ Cathy Wu's lab, MIT, Apr. 2025
- White-Box Transformers @ SVIP Lab, ShanghaiTech, Sep. 2023
- Emergence of Segmentation in White-Box Transformers @ CPAL 2024, HKU, Jan. 2024

Service

Head Teaching Assistant — HKU DATA 8014

Reviewer — ICLR 2025/2026, ICML 2025, NeurIPS 2025, CVPR 2026